

TRINAV CHAUDHURI

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Applied AI engineer. Production systems across clinical documentation, regulatory compliance, edtech, real estate, and neuroimaging.

EDUCATION

Cornell Tech – Jacobs Technion-Cornell Institute

Dual M.S. in Information Science, Health Tech Concentration

University of California, Berkeley

B.A., Double Major in Data Science and Molecular & Cellular Biology

New York, NY

Aug 2026 – May 2028

Berkeley, CA

Aug 2022 – May 2026

EXPERIENCE

Omi Health

AI/ML Engineer

Feb 2026 – Present

New York, NY

- Owned a clinical planning subsystem end to end, from schema and a deterministic assignment engine through LLM generation, validation, API, and migrations: 10,502 lines across 26 modules, roughly two thirds of the commits, row-level security.
- Consolidated an 11-call LLM fan-out into one structured call, decided by eval: duplicate outputs 7 to 0, coherence up, 34% fewer tokens, quality-neutral; adopted as the production default after review by both technical leads.
- Chose determinism over model confidence for generation reliability: deterministic logic gates every output and the LLM advises but never gates; the pattern resolved a senior engineer's objection to an LLM mid-pipeline and became the team standard.
- Built and gated the team's autonomous coding-agent workflow: isolated worktrees per task, explicit exit conditions, and automated review gates every change clears before merge, so agent-authored work meets the same bar as hand-written; designed the test coverage strategy validating the subsystem.
- Root-caused an 8-week silent retrieval outage that logged nothing (a circular import frozen to None plus a misread config accessor), consolidated two divergent implementations into one typed service with hybrid graph, vector, and fulltext retrieval, and replaced a 120s hang with a 15s-timeout fallback, measured by a 27-case eval retaining 100% emergency detection.

Ji Lab, UC Berkeley Physics

Research Assistant

Aug 2023 – Dec 2025

Berkeley, CA

- Found and fixed four defects in a voltage-imaging pipeline under principal-investigator review, correcting the dF/F baseline from -40% excursions to centered; awarded the *Hearts to Humanity Eternal Grant* (1 of 12 annually).

Magna Education

Data Science Intern

Jun 2025 – Aug 2025

San Francisco, CA

- Built a cheating-detection system over quiz answers: Azure OpenAI embeddings, cosine-similarity clustering, and PCA/t-SNE over a labeled benchmark of GPT-4o, Gemini, Claude, and human-written answers.
- Ran the adopt-or-not vendor evaluation of Copyleaks and ZeroGPT against that benchmark and designed the thresholds and short-answer guardrail against false positives.

Blackprint AI

Data Science Intern

Aug 2024 – Jan 2025

San Francisco, CA

- Built an async BFS crawler over three listing sites and four Mexican cities with proxy rotation, reconciling 400K+ listings into one schema.
- Fed incremental S3 loads with on/off-market state diffing so only changed records reprocessed, with downstream cleaning on AWS Redshift using complex SQL (joins, unions, geospatial functions).

PROJECTS

RegWatch – Automated Compliance Remediation

Grand Prize, Toolhouse x 500 Global Hackathon

- Led a 2-person team (34 of 37 commits) building a 4-agent pipeline (scraper, analysis, impact, remediation) that auto-generates fix PRs for HIPAA, SOC 2, GDPR, and FDA 21 CFR Part 11 violations; 60% auto-fix rate on HIPAA violations across three hospital test codebases scoring 20, 67, and 95.
- Designed for regression, not just generation: AST-based compliance checkers regenerated from versioned prompts across 12 modules, with the test suite accumulating from 47 to 51 checks across a regulation change so prior coverage survives.

AreWeThrough – World Cup 2026 Qualification Tracker

arewethrough.com

- Built a knockout-qualification tracker on a two-source verification pipeline that refuses to publish scores its independent sources dispute: published results are immutable, and disagreements open alerts instead of silent updates.
- Wrote the 561-test suite first, including 495 table-driven cases mechanically extracted from the official regulations, and verified the combinatorial scenario engine against an independent 100k-run Monte Carlo within one sigma.
- Ran a forced-mismatch drill against the live pipeline rather than trusting it, surfacing a timezone date-bucketing bug affecting a third of records.

EazyHealth – Automated Health Literacy Briefings

FastAPI, PostgreSQL, React; Render + Vercel

- Built an LLM abstraction layer over Anthropic and OpenAI with tiktoken budgeting, generating briefings at five reading levels from grade 3 to college, plus a duplicate checker scoring each new topic against a 30-day window on weighted tag and title similarity.
- Automated the loop end to end on a weekday scheduler that rotates 89 health topics through a 4-week reading-level cycle.

Retracing Intelligence – Evolution of Intelligence as Stacked Agents

playground

- Simulated the mechanisms in *A Brief History of Intelligence* that were hardest to picture from the text, written up alongside reading notes: Layer 1 of five, a dependency-free 2D simulation reproducing six named neuroscience experiments from the *C. elegans* copper test to Berridge's wanting/liking dissociation.
- Designed a dual-channel reward architecture separating wanting from liking as a structural answer to reward hacking and never-satisfied agents in single-scalar RL.

SKILLS

Languages Python, SQL, TypeScript/JavaScript, MATLAB

ML & AI LLM orchestration, multi-agent pipelines, evals (LLM-as-judge, regression suites), RAG/hybrid retrieval, prompt engineering, scikit-learn

Backend FastAPI, Flask, Docker, asyncio, Next.js, React, pandas, NumPy/SciPy, GitHub Actions, Git, Linux

Data & Cloud PostgreSQL, Supabase, AWS (S3, Redshift), Azure OpenAI, Render, Vercel

ATHLETICS & ARTS

USA Karate national team, Pan-American Championships (Brazil, Ecuador), world-ranked #9 (2019) | 18 years of tabla.